DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

IT25-425 P-II
COMPUTER NETWORKS
(MORNING & EVENING)

LAB#1

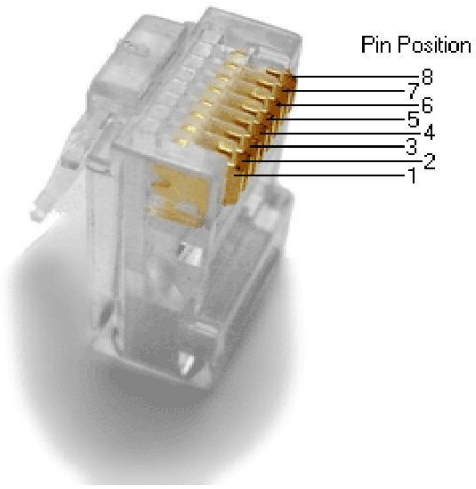
TASK: UTP CABLE CONSTRUCTIION

Aim: Study of different types of Network cables and Practically implement the cross-wiredcable and straight through cable using clamping tool.

Apparatus (Components): RJ-45 connector, Crimping Tool, Twisted pair Cable Cat-5e or 6

Cabling Information:

Explanation: Instructions are provided here for building a T568-A or T568-B cable. Either can be used as long as all connections (pinouts) from the workstation to the wiring closet and terminating electronics (hubs or switches) are consistent. If cables are to be built for an existing network it is important to keep the same standard as already exists (either T568-A or B). A patch cable that is wired "straight through" will have the same color of wire on the same pin (1 -- 8) at both ends. A straight through patch cable (T568-A or B) can be used to connect a PC workstation to a wall plate in a work area or it can be used to connect from a patch panel in a wiring closet to a hub or a switch. A PC can also be connected directly to a port on a hub or switch with this cable. If a cable will be used to connect from an "uplink" port on one hub to a "crossover" front port on another hub then a straight through cable should be used





Pin #	Pair#	Function	Wire Color	Used with 10/100/1000 BASE-T Ethernet
1	2	Transmit	White/Orange	Yes
2	2	Transmit	Orange/White	Yes
3	3	Receive	White/Green	Yes
4	1	Not used	Blue/White	No
5	1	Not used	White/Blue	No
6	3	Receive	Green/White	Yes
7	4	Not used	White/Brown	No
8	4	Not used	Brown/White	No



Diagram showing both T568-A and T568-B cabling wire colors

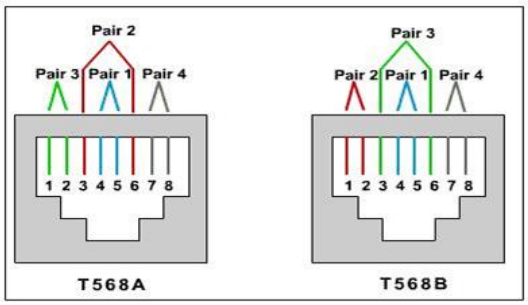
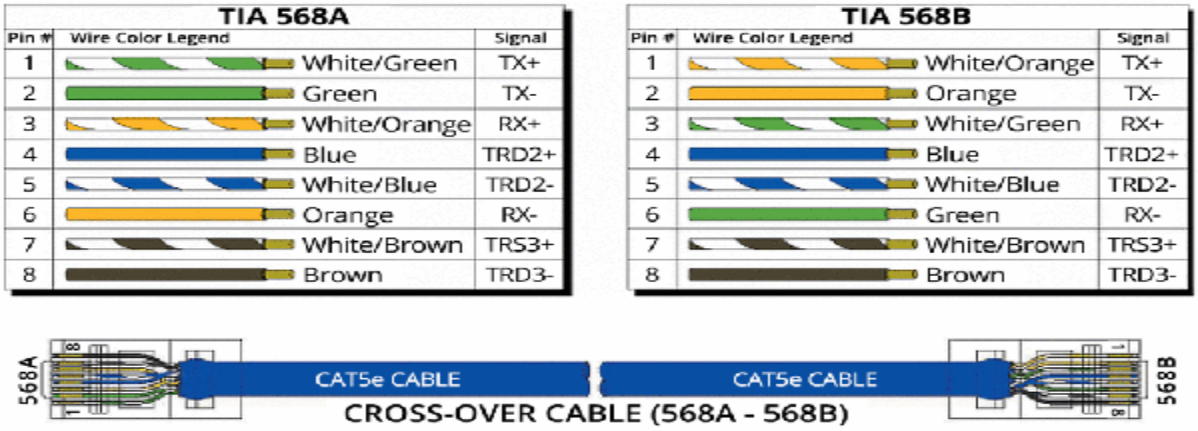


Diagram shows you how to prepare Cross wired connection



Procedure: To do these practical following steps should be done:

1. Determine the distance between devices, or device and plug, then add at least 12" to it. The maximum length for this cord is 3 m; standard lengths are 6' and 10'.

2. Cut a piece of stranded Cat 5 unshielded twisted-pair cable to the determined length. You will use stranded cable for patch cables because it is more durable when bent repeatedly. Solid wire is fine for cable runs that are punched down into jacks.

3. Strip 2" of jacket off of one end of the cable.

4. Hold the 4 pairs of twisted cables tightly where jacket was cut away, then reorganize the cable pairs into the order of the 568-B wiring standard. Take care to maintain the twists since this provides noise cancellation. (orange pair, green pair, blue pair, brown pair).

5. Hold the jacket and cable in one hand, untwist a short length of the green and blue pairs, and reorder them to reflect the 568-B wiring color scheme. Untwist and order the rest of the wire pairs according to the color scheme.

6. Flatten, straighten, and line up the wires, then trim them in a straight line to within 1/2" - 3/4" from the edge of the jacket. Be sure not to let go of the jacket and the wires, which are now in order! You

should minimize the length of untwisted wires because overly-long sections that are near connectors are a primary source of electrical noise.

7. Place an RJ-45 plug on the end of the cable, with the prong on the underside and the orange pair to the left side of the connector.

8. Gently push the plug onto wires until you can see the copper ends of the wires through the end of the plug. Make sure the end of the jacket is inside the plug and all wires are in the correct order. If the jacket is not inside the plug, it will not be properly strain relieved and will eventually cause problems. If everything is correct, crimp the plug hard enough to force the contacts through the insulation on the wires, thus completing the conducting path.

9. Repeat steps 3-8 to terminate the other end of the cable, using the same scheme to finish the straight through cable.

10. Test the finished cable and have the instructor check it. How can you tell if your cable is functioning properly?

Review Questions

Question 1: Use straight through Ethernet cable for the following cabling/devices
Answer:

_____
_____

_____

Question 2: Use crossover cables for the following cabling
Answer:

_____
_____
_____

Note: Submit the handout before next lab